SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	M.V.Prasanth	**Reg. No.:**	192472219
Section / Batch:	CSE AI	**Date:**	13/07/2026

Code of Conduct

I M.V.Prasanth(192472219)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ____________________________

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.
Identify email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
Extract years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

Section 1) Aim:

To extract candidate details from a resume using Regular Expressions and check eligibility.

Procedure:

- **Import the re module.**
- **Enter the resume text.**
- **Extract the name, email, phone number, experience, and skills using Regular Expressions.**
- **Display the extracted details.**
- **Check the eligibility criteria (minimum 2 years of experience and Python skill).**
- **Display Eligible or Not Eligible.**
- **Stop.**

Code:

```
import re resume
= """
```

```
Name: Rahul Sharma

Email: rahul@gmail.com

Phone: 9876543210

Skills: Python, Java, SQL

Experience: 3 years

"""


name = re.search(r"Name:\s*(.*)", resume).group(1) email =

re.search(r"[\w\.-]+@[\w\.-]+\.\w+", resume).group() phone

= re.search(r"\d{10}", resume).group()

exp = int(re.search(r"(\d+)\s+years", resume).group(1))


skills = [] for s in ["Python", "Java", "SQL", "Machine Learning",

"NLP"]:        if  re.search(s,  resume,  re.IGNORECASE):

skills.append(s)


print("Name:", name) print("Email:",

email) print("Phone:", phone)

print("Experience:", exp, "years")

print("Skills:", skills)


if exp >= 2 and "Python" in skills:

   print("Eligible") else:

   print("Not Eligible")
```

Output:

Name: Rahul Sharma

Email: rahul@gmail.com

Phone: 9876543210

Experience: 3 years

Skills: ['Python', 'Java', 'SQL']

Eligible

Result:

The program successfully extracts resume details using Regular Expressions and checks whether the candidate is eligible based on experience and Python skill.

Section 2) Aim:

To implement a product search system using Regular Expressions for searching products with different search patterns.

Procedure:

- Import the re module.
- Create a list of product names.
- Accept a search keyword.
- Perform exact, prefix, suffix, partial, and case-insensitive searches.
- Display the matching product names.
- Display the total number of matching products.

Code:

```
import re products
= [
  "Python Book",
```

```
    "Java Book",
    "SQL Guide",
    "Python Basics",
    "Machine Learning",
    "NLP Toolkit",
    "Python Notebook"
]
keyword = "Python"

exact = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)] prefix
= [p for p in products if re.match(keyword, p, re.IGNORECASE)] suffix = [p for
p in products if re.search(keyword + r"$", p, re.IGNORECASE)] partial = [p for
p in products if re.search(keyword, p, re.IGNORECASE)] print("Exact Search:",
exact) print("Prefix Search:", prefix) print("Suffix Search:", suffix)
print("Partial Search:", partial) print("\nReport")
print("Exact Matches:", len(exact)) print("Prefix
Matches:", len(prefix)) print("Suffix Matches:",
len(suffix)) print("Partial Matches:",
len(partial))
```

Output:

Exact Search: []

Prefix Search: ['Python Book', 'Python Basics', 'Python Notebook']

Suffix Search: []

Partial Search: ['Python Book', 'Python Basics', 'Python Notebook']

Report

Exact Matches: 0

Prefix Matches: 3

Suffix Matches: 0 Partial

Matches: 3
Result:

The program successfully performs exact, prefix, suffix, partial, and case-insensitive product searches using Regular Expressions and displays the matching products with the total number of matches.

Section 3) Aim:

To validate student registration details using Regular Expressions and display the registration status.

Procedure:

- **Import the re module.**
- **Enter the student details.**
- **Validate the register number.**
- **Validate the institutional email.**
- **Validate the course code.**
- **Validate the semester.**
- **Validate the mobile number.**
- **Display validation messages.**
- **Display the final registration status.**

Code:

import re regno = "192472215"

email =

"student@saveetha.com" course

```
= "CSE101" semester = "5"
mobile = "9876543210"


valid = True


if re.fullmatch(r"\d{9}", regno):
print("Register Number: Valid") else:
    print("Register Number: Invalid")
valid = False


if re.fullmatch(r"[\w\.-]+@saveetha\.com", email):
    print("Email: Valid") else:
    print("Email: Invalid")
valid = False


if re.fullmatch(r"[A-Z]{3}\d{3}", course):
    print("Course Code: Valid") else:
    print("Course Code: Invalid")
valid = False


if re.fullmatch(r"[1-8]", semester):
    print("Semester: Valid") else:
    print("Semester: Invalid")
valid = False
```

```
if re.fullmatch(r"\d{10}", mobile):
print("Mobile Number: Valid") else:
    print("Mobile Number: Invalid")
    valid = False
```

if valid:

print("Registration Successful") else:

print("Registration Failed")

Output:

Register Number: Valid

Criteria	Weightag e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandin g of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts

ProblemSolving Approach	20	Logical, wellstructured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

Email: Valid

Course Code: Valid

Semester: Valid

Mobile Number: Valid

Registration Successful

Result:

The program successfully validates the student registration details using Regular Expressions and displays the final registration status.

Rubrics for Calculation

Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, wellorganized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	ProblemSolving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director

Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________